Critical Study Summary: Simões et al. (2023) Pooled Nirsevimab Efficacy Analysis

Study Design and Population

Prespecified pooled analysis of two randomized, double-blind, placebo-controlled trials evaluating single-dose nirsevimab in preterm and term infants. Infants were randomized 2:1 to nirsevimab or placebo.

Follow-up and Time Scale

Infants were followed from dosing through 150 days post-dose, corresponding to a typical RSV season. Time-to-first RSV event was the principal analytical scale.

Primary Outcomes and Effect Measures

Primary endpoint was medically attended RSV lower respiratory tract infection. Effect measures included relative risk reduction and supportive hazard ratios from Cox proportional hazards models.

Key Findings Relevant to Waning

Protection against medically attended RSV-LRTI was sustained across the 150-day follow-up period, with hazard ratios indicating strong and persistent efficacy throughout the season.

Graphical Data for Reconstruction

Kaplan–Meier survival curves with numbers at risk and censoring indicators are provided for medically attended RSV-LRTI and RSV-related hospitalization, making the study well-suited for IPD reconstruction.

Strengths for Waning Analysis

Explicit time-to-event framework, availability of hazard ratios, and clear definition of censoring enhance suitability for advanced waning analyses.

Limitations for Waning Analysis

Follow-up limited to 150 days restricts inference to within-season waning; pooled analysis may mask heterogeneity between gestational age subgroups.